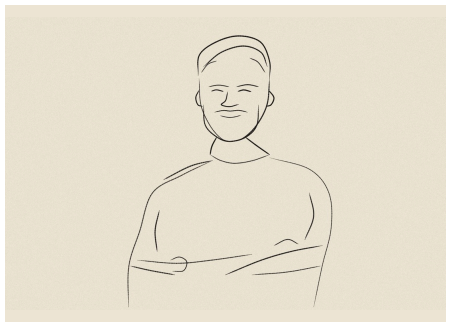
Sketch Lab: Constrained Line Drawing with a Vision-Language Model

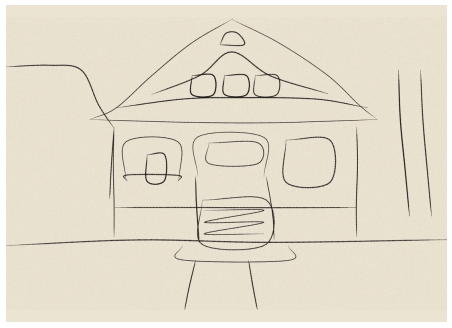
A drawing system in which a language model studies a photograph, plans a line drawing and makes it within a fixed number of pencil strokes

Dan Bartlett · *Independent Experimental Research · 2026*



PORTRAIT · BUDGET 30

Hair, beard, smile and crossed arms. The people in the background are left out.



HOUSE · BUDGET 30

Gable, three upper windows, porch, steps and path. Straight edges come out rounded.



TREE · BUDGET 30

The horizon and trunk hold. The canopy becomes a set of sweeping lines.



MOTORCYCLE · BUDGET 30

Strokes spent on the building behind it, and wheels drawn several times over.

Figure 1 — *Four drawings from the first test session, each limited to 30 strokes. Every line was placed by the model; the system only smooths and draws it.*

ABSTRACT

Sketch Lab is a small drawing system. You give it a photograph and a number of strokes, and a vision-language model decides how to draw the image within that limit. The drawing appears on screen stroke by stroke, with a label for each mark.

The model does not trace the photograph. Before it places any line, it is asked to describe what it sees, fix a set of landmarks on the subject and write out a plan for every stroke. Only then does it give the coordinates. A separate part of the system turns those coordinates into tapered pencil lines on a sheet of paper.

The stroke limit is the main control, and it can be set anywhere from 3 to 120. With fewer strokes, the model has to decide what matters most in the image. **What it leaves out tells us something about how it reads a picture.**

MOTIVATION

Drawing is largely a matter of deciding what to leave out. A quick sketch from life records what the person drawing thought was important, and two people drawing the same subject will rarely agree. I wanted to see what happens when that decision is handed to a machine, under the kind of pressure a short pose puts on a student in a drawing class.

Sketch Lab is also an example of system co-creation, the second mode of practice in my research. I wrote the constraints, the prompt and the renderer, working with Claude as a coding assistant. The model makes the drawing decisions. I chose the photographs and the budgets, and I am still working out how to describe my part in the drawings that result.

HOW A DRAWING IS MADE

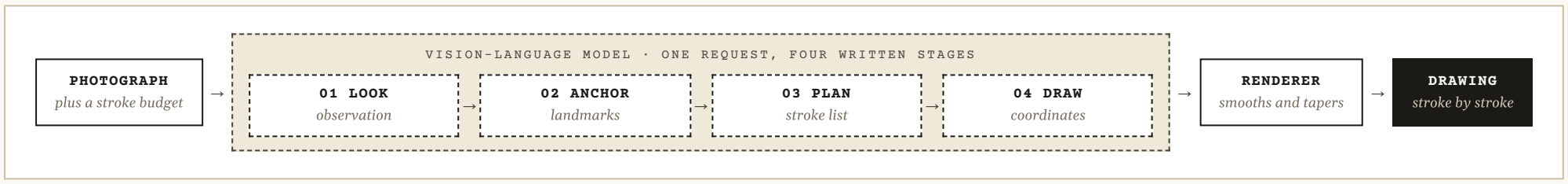


Figure 2 — *The system. The model's decisions sit inside the dashed area. Everything after it is fixed procedure, apart from a small random wobble and variation in line weight that make each stroke look hand-drawn.*

The prompt asks the model to work through four stages in order and to write each one down before moving on. Nothing in the system forces that order; the prompt simply asks for it, and the written stages are shown alongside the finished drawing.

01
LOOK

Two or three sentences on the subject, the mood and the light.

Shown under the drawing.

02
ANCHOR

Eight to fifteen named landmarks with coordinates, such as the corner of an eye, the peak of a roof or the centre of a wheel.

Can be laid over the drawing.

03
PLAN

A numbered list of what each stroke will draw: outline first, then large forms, then features, then details, in proportion to the budget.

At 10 strokes: 1, 3, 4 and 2.

04
DRAW

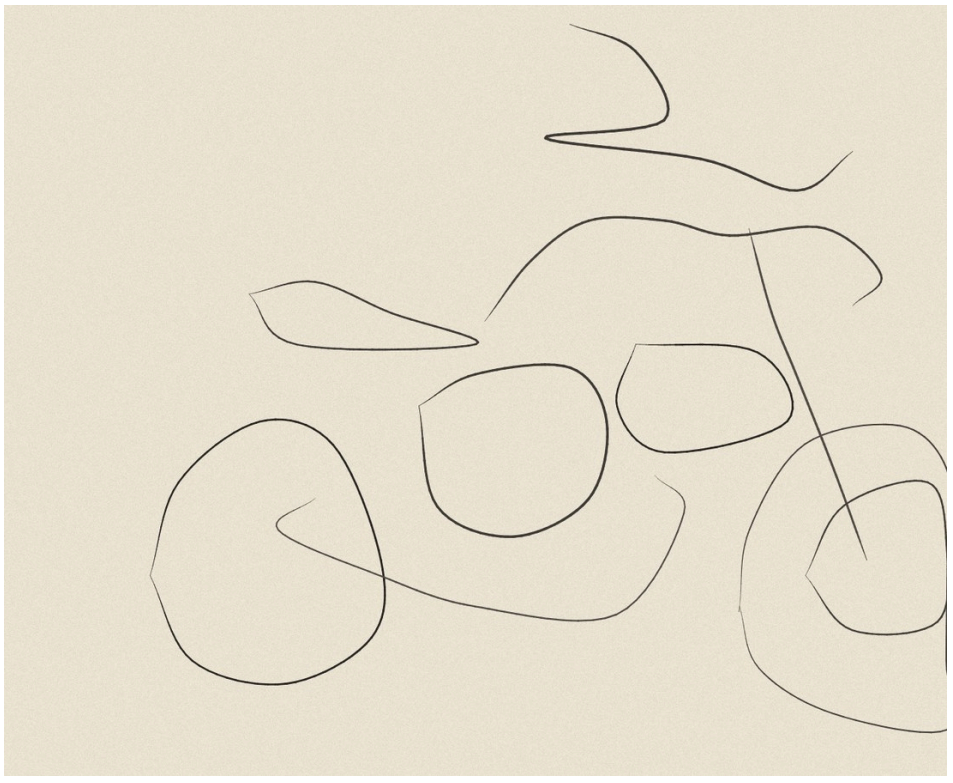
Each stroke as a short list of points. The renderer smooths these into a curve and draws a line that thickens and tapers like pencil.

Each stroke is labelled as it appears.

If a machine can make only ten marks,
which ones does it choose?



BUDGET 30



BUDGET 10

Figure 3 — *The same photograph at two budgets. At 30 strokes the model spent long strokes on the building behind the motorcycle and drew each wheel several times. At 10 it left out the background and gave one or two strokes to each main part: the wheels, tank, seat, engine, fork and handlebars.*

WHAT THE FIRST DRAWINGS SHOW

These come from one short session: four photographs at 30 strokes and one of them again at 10. One drawing per subject is not enough to generalise from, so I treat what follows as things to test.

SUBJECTS

The portrait was the most successful drawing. The hair, beard, smile, collar and crossed arms are all there, and the two people working in the background were left out entirely. Earlier versions of this poster described portraits as the hardest subject for the system. This drawing does not support that.

The house kept its structure: the gable, the three upper windows, the porch, the steps and the path. Its straight edges came out rounded. That comes from the renderer, which smooths every stroke into a curve, and it is the next thing I intend to fix.

The tree and the motorcycle were weaker. The canopy has texture but no clear outline, and it became a set of sweeping lines. The motorcycle is a dense arrangement of overlapping parts, and at 30 strokes it became a tangle.

BUDGETS

Cutting the motorcycle from 30 strokes to 10 made it clearer (Figure 3). With fewer strokes available, the model stopped drawing the background and stopped repeating itself.

This is the effect the stroke limit was designed to produce. It suggests that a tighter limit may help with subjects that the model struggles with at higher budgets. I want to test that with the tree, and with other dense subjects, before relying on it.

The drawings are also a record of choices I did not make. I chose the photographs and the budgets. What to draw, what to keep and what to leave out were decided by the model, and its observation, landmarks and plan show its working.

WHO IS DRAWING?

When I set the budget to 10 and the motorcycle became legible, I had not told the system what to remove. I had only allowed fewer marks, and the model decided how to use them. To me that feels closer to setting an exercise for a student than to operating a tool.

It also connects to the experience at the centre of my research: working with a system that does not simply carry out what I ask, and that sometimes produces something better because of the limits I set.

THE OPPOSING VIEW

There is a simpler explanation. The model has seen a great deal of writing about drawing, and it produces text that resembles a drawing plan because of that. On this reading, no judgement is involved. The landmarks and the plan describe the output without explaining how it was reached.

I take that view seriously, and the drawings cannot settle the question. They are useful as something concrete for practitioners to look at while it is argued.

WHAT IT DOES NOW

- Draws from any photograph that is dropped, pasted or chosen
- Stroke budget from 3 to 120, with the plan scaled to fit
- Five alternative approaches to the same image through REDRAW
- Shows the model's observation, landmarks and full stroke plan
- Saves drawings as PNG or editable SVG, and the analysis as JSON
- Logs every run, with timings, for later analysis

NEXT STEPS

- Run the tree and other dense subjects at several budgets
- Repeat each test to see how much the drawings vary between runs
- Keep sharp corners on straight-edged subjects
- Try the same prompt on a local open-weight model (Qwen2.5-VL)
- Compare the model's choices with drawings made by students working to the same budget